Coffee Shop Analysis: Advertisement and Marketing Strategies by Season

Disclaimer and Limitations

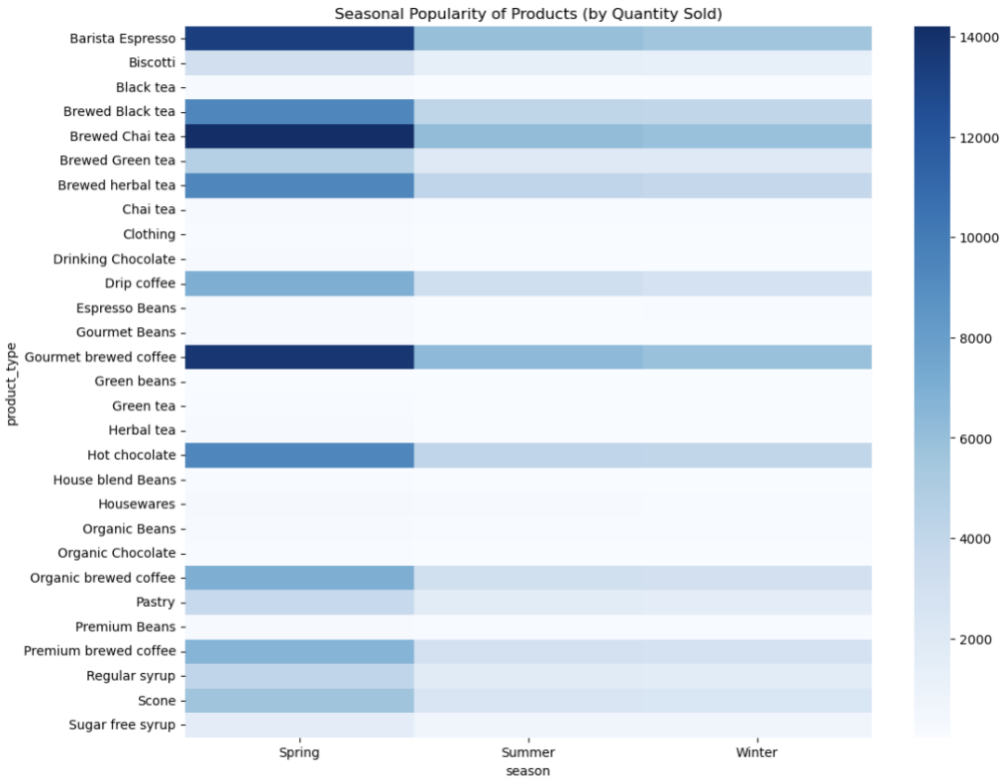
The following data is obtained from a coffee shop located in NYC. The data only includes information for the first six months of 2023, limiting our ability to find seasonal patterns conclusively. This can cause imbalance in the data, and provide limitations to our insights. This data does not include the fall season, thus only includes winter, summer, and spring. Therefore, spring having more transaction information than the rest of the months did cause a slight skewness in the distribution of our data. It is important to disclose limitations in data, since it is not always perfect, but can still give us ambitious insights and valuable information.

Model used for predictions:

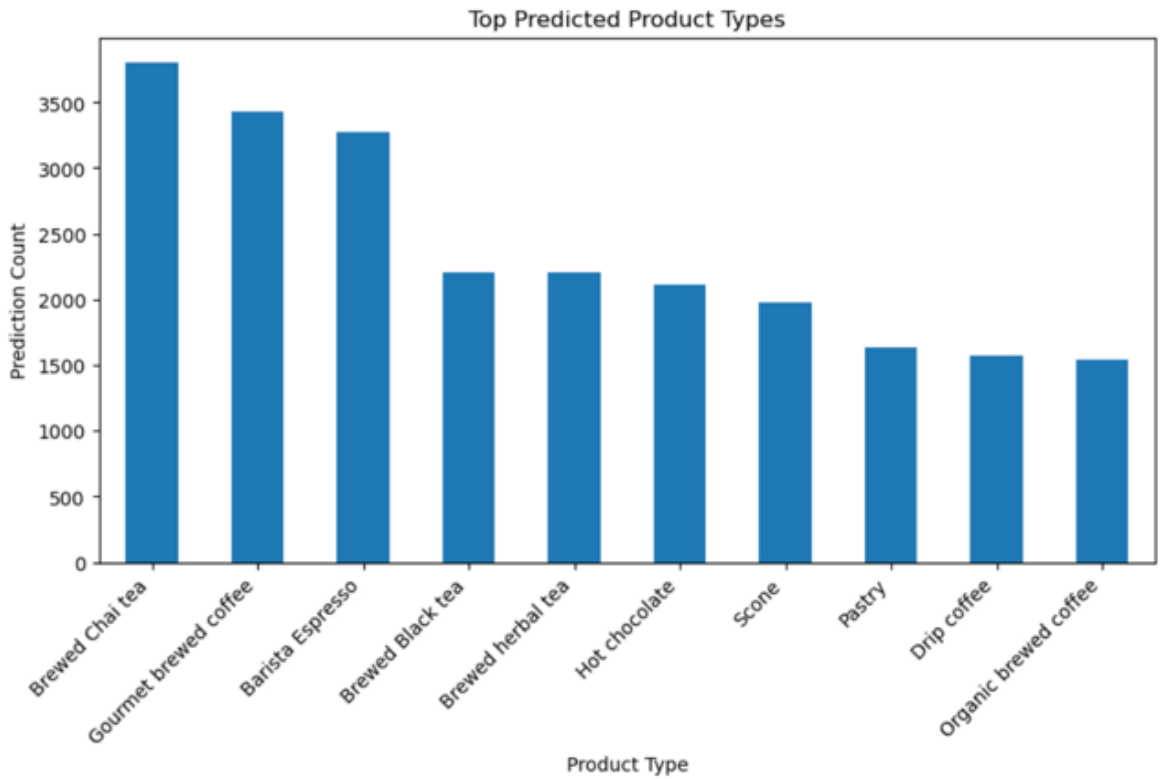
The models used for making coffee shop revenue and sales prediction analysis were multi class logistic regression and random forest classifier. The best performing model was random forest classifier, with a stunning accuracy of 91.8%. This model can be used by the coffee shop for future data and have a better understanding of revenue generating products before the season starts. This will help them get to know their customer's taste, which evolves every season!

Visual of predictions:

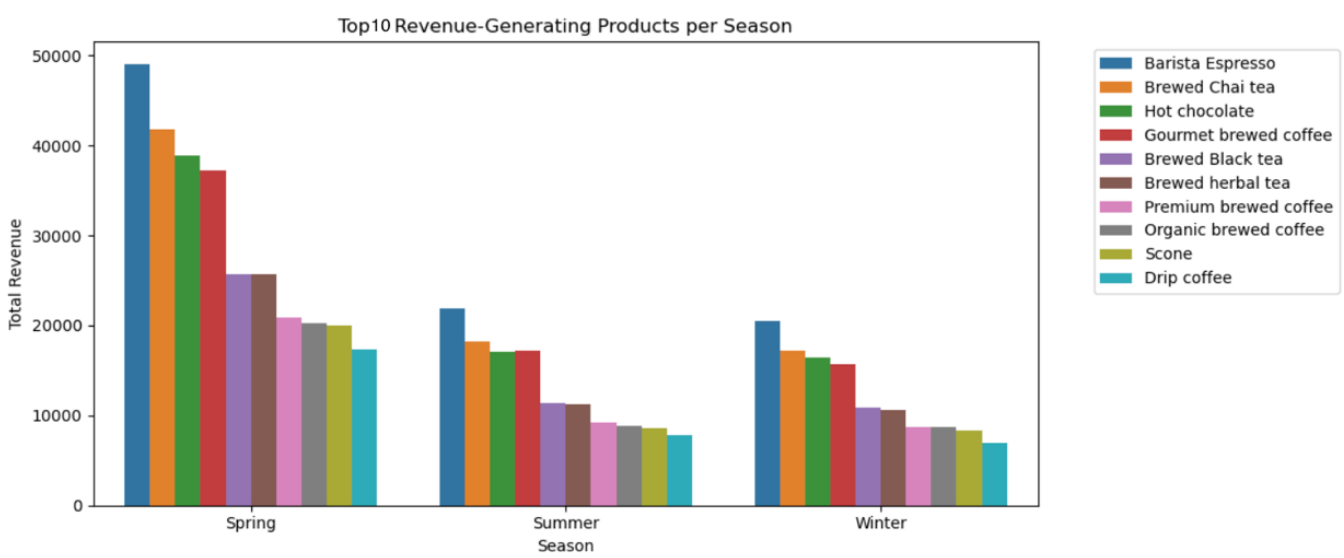
Heatmap



Bar Graph (Top Predicted Products)



Bar Graph 2 (Top Products by season)



Strategies by Season

Spring

Advertisement boards posted outside stores always reel customers in and influence them to come pay a visit inside. The most revenue generating product is brewed chai tea (after the barista espresso, which is used in many coffee drinks such as latte, cappuccino, etc.) an advertisement for brewed chai tea would be a good strategy to encourage customers to come in and try the seasonal favorite product. Hot chocolate is also comparably higher than the other season, and can be displayed outside on the colder/ rainier spring days.

Compared to other seasons, spring has the most generated revenue from pastries like scones. An effective marketing strategy could be used to combine this with the most popular drinks such as chai tea and drinks that have barista espresso in them. These include but are not limited to: Lattes, Cappuccinos, Flat Whites, Mochas and Americanos. A scone and drink combo promotion can be used for the certain times of the day (typically when it is slow). A simple sign outside the café like *"Get a cranberry scone with your Chai tea, and get a taste of spring"* can encourage customers to step in, leading to more revenue being generated through the favored products and maintaining an increased presence of customers inside the café during slow hours.

Summer

A good advertisement strategy for summer would be to display brewed black coffee on a sign outside the store. Since all seasons are typically following a similar trend, we have to look at slight differences between seasons. Compared to winter season, summer season predicts a slightly higher revenue for brewed black coffee. This can be served on ice or as cold brew. Iced coffees displayed outside the store would encourage customers to come in and cool of from the hot sun, all while refreshing and caffeinating themselves.

Winter

During the winter, warmer drinks are proffered, and the data shows hot chocolate and warm drinks that use espresso is a good strategy to advertise outside the store. Using words like "rich" "cozy" and "seasonal", can help with bringing in more customers.

As for marketing strategies during the colder season, it can be hard since people are most likely not leaving their space and walking around. Partnering with local events around the city can help boost visibility of the café, and providing samples of the warm drinks can help boost brand awareness during a time where people prefer to be spending time with family inside and staying cozy. Winter Farmer markets, skating rinks, or holiday fairs is a great strategy to meet your customers where they are. As for in store marketing strategies, offering 'limited edition" products is a great technique to generate revenue. The holidays is a great time to introduce products such as "peppermint hot chocolate" and "gingerbread latte", classic warm winter delights that cannot be ignored by most. This successful strategy can be seen practiced by the Starbuck's corporation, generating over between $400 million and $800 million a year, with $500 million being a "decent" estimate.

Sources/Citations:

- ***Passy, Charles, and Getty Images. 2024****. "Starbucks Will Likely Make Over $500 Million on Pumpkin-spice Lattes This Season. Is That Enough?" MarketWatch, August 26, 2024.* *https://www.marketwatch.com/story/starbucks-pumpkin-spice-latte-could-be-a-500-million-seller-but-is-that-good-enough-04cfccc8?gaa_at=eafs&gaa_n=AWEtsqfUL9EYA_wY9eAKfe-sbdcmsy4zF1AL53BLOtISlqop2j8XEzCvnM5w7cdBakg%3D&gaa_ts=69388ffd&gaa_sig=_6tyS-nQhoRT4F*